

Marwan Aljijakli

BRIEF INTRODUCTION

AI & Data Engineer with end-to-end technical ownership of data pipelines, model inference, secure APIs, testing, and cloud deployment. Leads BOHIO's architecture for AI-assisted financial modeling and PostgreSQL/Supabase workflows. At VLEED, builds Dockerized rPPG services and GPU-accelerated computer vision and generative AI pipelines using Python, PyTorch, FastAPI, and OpenCV.

PERSONAL INFO

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Saudi Council of Engineers (SCE) - Active Membership No. 1272601 | Valid through 13 Jul 2027

EXPERIENCE

BOHIO - Chief Technology Officer (Hands-on Technical Lead) | Jan 2026 - Present

- Lead architecture and hands-on delivery of a real estate financial modeling platform with AI-assisted workflows, Next.js, TypeScript, Supabase/PostgreSQL, ExcelJS, and HyperFormula.
- Engineered its calculation engine, APIs, collaboration, billing, bilingual experience, administration console, row-level security, telemetry, tests, and deployment controls.

VLEED - AI Engineer | Jun 2024 – Dec 2025

- Productized an open source rPPG/POS pipeline as a Dockerized FastAPI service using OpenCV, LinkNet segmentation, and FFT signal analysis to estimate heart rate from facial video.
- Built FacePic workflows with InsightFace, InstantID, SDXL, and Stable Video Diffusion; contributed to a Flutter health app and Flask/MySQL backend.

SELECTED PROJECTS

Connected IoT Monitoring Systems - Personal Project Portfolio | 2023 - 2025

- Built ESP32/MQTT telemetry, embedded firmware, BLE workflows, React/Chart.js dashboards, REST integrations, device diagnostics, edge AI prototypes, and OTA firmware.

STM32 Sensor & Telemetry Platform - Personal Project | 2024

- Developed STM32 HAL drivers, FreeRTOS/I2C/SPI integration, Python/InfluxDB ingestion, Grafana dashboards, and hardware validation with oscilloscopes, logic analyzers, and JTAG.

Wathba - Personal Edge and Cloud Monitoring Project

- Built ESP32 firmware, NVS configuration, authenticated HTTPS telemetry to Supabase/Vercel, and dashboard flows for stale/offline detection and device diagnostics.

SKILLS

- AI, Computer Vision & Signal Processing:** Python, PyTorch, TorchVision, OpenCV, NumPy, SciPy, rPPG/POS, FFT, semantic segmentation, LinkNet, image/video processing
- Generative AI & GPU Inference:** Hugging Face Diffusers, InsightFace, InstantID, ControlNet/IP-Adapter, SDXL, Stable Video Diffusion, ONNX Runtime GPU, FP16/TF32, VRAM optimization, CPU/model offload, OOM recovery
- LLM & AI Application Systems:** Anthropic/OpenAI APIs, Vercel AI SDK, streaming, tool calling, structured output validation, provider fallback, rate limiting
- Data & Backend Engineering:** FastAPI, Flask, REST APIs, SQL, PostgreSQL, Supabase, MySQL, Python ETL, data ingestion and validation, schema design, database migrations, RLS, authentication
- Cloud, Quality & Product Delivery:** Docker, Docker Compose, Vercel, GitHub Actions CI, Sentry/telemetry, health checks, Playwright, Vitest, Next.js, TypeScript, React, Flutter/Dart

AWARDS:

1st Place - SAQIR Saudi Aerial Quadcopter Engineering Race (Drone Challenge) | 2025

1st Place - CPC Programming Problem-Solving Competition, Jeddah International College | Team Award

3rd Place in Saudi Arabia - Taghna AI Hackathon Project | 2025

Education & Registration

- Bachelor of Science in Computer Science:** Specialization in Artificial Intelligence - Jeddah International College | 2026 | GPA: 4.35/5.0
- Applied AI Training:** SDAIA advanced AI/ML curriculum and projects; KAUST applied AI research and engineering program
- Leadership:** President, SWAICY Club; Head of Student Affairs, Computer Science/IT